BLOCKCHAIN · NFT · DECENTRALIZED SECURITY

BLOCK TICKET

Immutable · Verified · Tamper-Proof Event Ticketing on the Blockchain

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Role:

AI & Data Science Students · Full-Stack Developers

Institution:

Rajalakshmi Engineering College · 2026

#000 GENESIS

#001 MINT NFT

#002 TRANSFER

#003 VERIFY

The 'Fake Ticket' Crisis

$12B+
Annual revenue lost to ticket fraud globally

~30%
Of resold tickets are counterfeit or invalid

68%
Fans report distrust in secondary markets

4.6M
Fraudulent tickets sold in the last 3 years

🎫 Counterfeiting & Scalping

- High-resolution printers replicate barcodes in minutes — standard scanners cannot detect fakes
- Bot networks sweep inventory within seconds of release, inflating resale prices by 300–1000%
- No on-chain provenance — impossible to verify original purchase without centralized records
- Single point of failure: if the ticketing platform's DB is breached, all tickets are compromised

🏢 Centralized System Failures

- Siloed databases provide zero transparency into ticket transfer history or ownership chain
- Organizers lack royalty mechanisms — revenue from resales goes entirely to scalpers
- Screenshot-based QR codes allow unlimited copies to be shared and reused at entry gates
- Venue management depends on a single vendor, creating monopoly pricing & data lock-in

The ticketing industry demands a trustless, decentralized solution — the status quo is broken.

The Solution — Why Blockchain?

❌ Centralized (Old Way)

- Single corporate database — one breach = all tickets invalid
- Barcodes are static & easily screenshot-copied by fraudsters
- Zero resale transparency; scalpers profit unchecked
- Organizer has no control once ticket leaves platform
- Buyer identity unverifiable — no KYC link to ticket

VS

✅ BlockTicket (New Way)

- ERC-721 NFTs on Ethereum/Polygon — immutable, decentralized ownership chain
- Time-based cryptographic QR codes — invalid 60 seconds after generation
- On-chain resale caps & creator royalties enforced by Smart Contracts
- Organizer retains royalty control for the full lifecycle of the ticket
- Wallet-based identity — every transfer logged on the public ledger

Why Blockchain? → Trustless, transparent, programmable, and censorship-resistant by design.

Core Objectives

NFT Ticket Minting
01

ERC-721 Standard

- Each ticket is a unique ERC-721 Non-Fungible Token minted on Polygon PoS for low gas fees
- Token metadata stored on IPFS: seat, event, tier, face value, and organizer signature
- Immutable provenance trail — every owner recorded from mint to gate scan
- Batch minting supported for large events (10K+ tickets) using gas-optimized loops

Smart Contract Price Caps
02

Anti-Scalping Logic

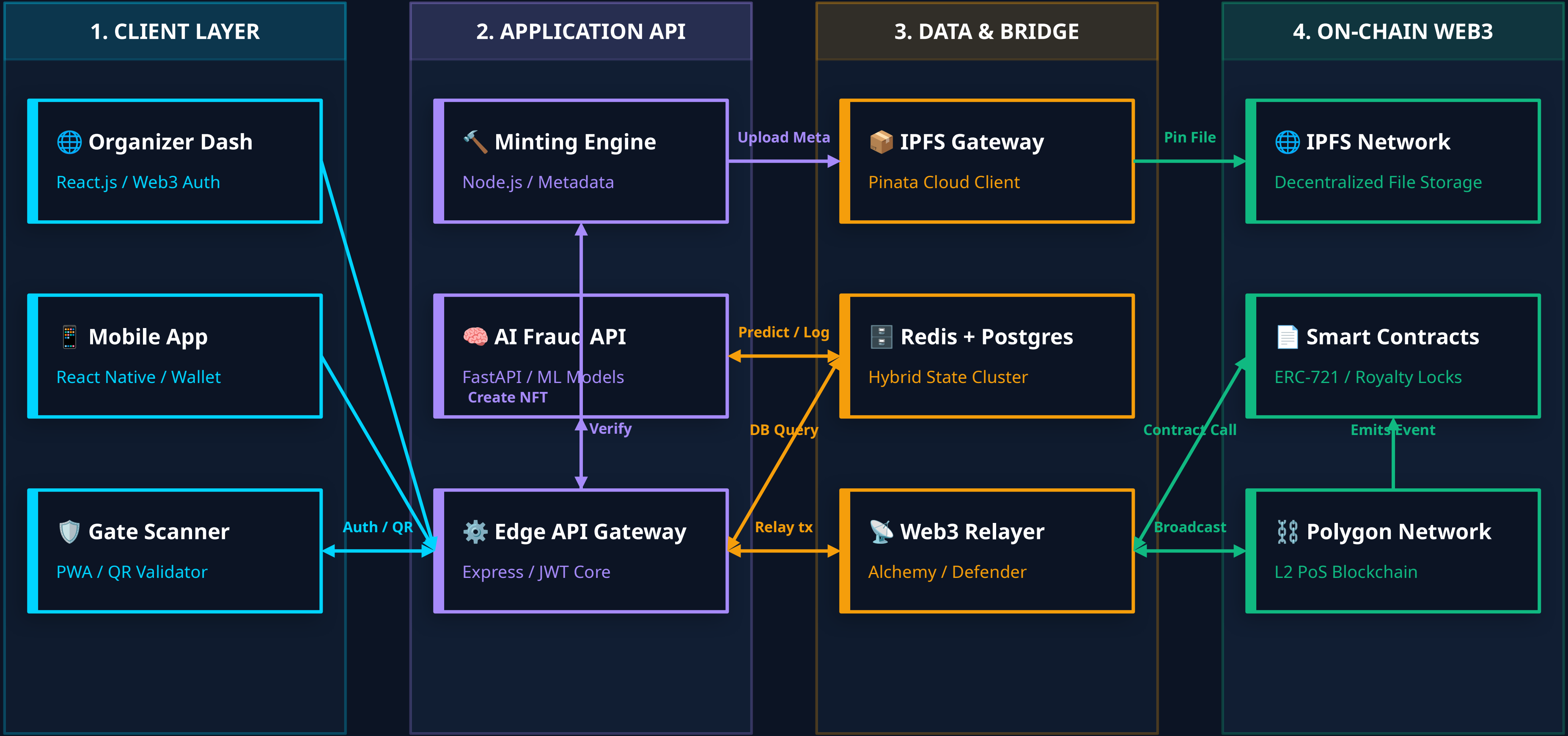
- Solidity-enforced resale cap: secondary price ≤ 150% of face value by default
- Creator royalty (5–15%) auto-transferred to organizer wallet on every resale
- Decentralized escrow: funds released only on verified gate scan event
- Price violation attempts revert on-chain — no off-chain workarounds possible

Dynamic QR Codes
03

Time-Based Auth Tokens

- TOTP-style QR generated fresh every 60 seconds using HMAC-SHA256 + NFT token ID
- QR payload encodes: wallet address, token ID, timestamp, and cryptographic signature
- Gate scanner validates signature against smart contract — screenshots are always expired
- One-time-use nonce prevents replay attacks even within the 60-second validity window

Industry Standard System Architecture



Key Features

💰 Royalty Enforcement Engine

For Organizers

- On-chain royalty hook in the ERC-721 `transferFrom` override — automatic, irrevocable
- Configurable royalty rate (0–20%) set at minting time; cannot be bypassed by any reseller
- Compatible with ERC-2981 Royalty Standard — works natively on OpenSea, Rarible, and custom marketplaces
- Royalty funds held in escrow until gate verification confirms attendee entry
- Real-time dashboard shows organizer how much royalty revenue is accumulating on-chain

🔒 One-Ticket · One-Owner Logic

For Trust

- Smart contract enforces a hard transfer lock after gate scan event is emitted — ticket becomes non-transferable
- Wallet-bound ownership: ticket NFT is permanently linked to the attendee's crypto address
- Transfer cooldown (configurable: 24–48 hrs before event) prevents last-minute scalping flips
- Blacklist mechanism: wallets flagged as bots or scalpers are blocklisted at the contract level
- Full audit trail: every ownership change is a traceable on-chain transaction with timestamp

Security Measures

⚠️ Screenshot Fraud

60-sec TOTP-style QR rotation via HMAC-SHA256(tokenID + wallet + timestamp). Expired tokens revert on-chain.

⚠️ Replay Attacks

One-time nonce per scan event recorded on-chain. Resubmitting a used nonce causes an immediate contract revert.

⚠️ Double Spending

Gate scan emits a `TicketUsed(tokenId)` event. Subsequent scans within the same block fail at the require() check.

🔐 Cryptographic QR Flow

① App calls generateQR(tokenId, walletAddress) every 60 s

② Server computes: HMAC-SHA256(secret || tokenId || floor(now/60))

③ QR payload = Base64(tokenId + walletAddress + hmac + nonce)

④ Gate scanner verifies HMAC, checks nonce registry, calls verifyTicket() on-chain

⑤ Smart contract emits TicketUsed(tokenId, scanner, timestamp)

🛡️ Additional Security Layers

- TLS 1.3 encryption on all API endpoints; no plaintext ticket data in transit
- Gas limit caps on smart contract functions — DoS attack surface minimized
- OpenZeppelin AccessControl for admin functions (MINTER_ROLE, PAUSER_ROLE)
- Chainlink VRF for tamper-proof randomized seat assignment on mint
- Multi-sig (2-of-3) Gnosis Safe for contract upgrade & treasury operations

Expected Impact

~100%
Counterfeit Elimination

5–15%
Organizer Royalty Per Resale

<150%
Resale Price Cap Enforced

60 sec
QR Validity Window

🏢 For Event Organizers

- Guaranteed royalty income on every secondary sale — zero dependence on ticketing platform
- Full visibility into resale velocity, demand spikes, and geographic attendee distribution
- Programmable transfer restrictions (blackout periods, geographic locks) via smart contract
- Reduced fraud investigation costs — blockchain provides irrefutable proof of authenticity
- Enhanced brand trust: verified ticket = premium experience reputation for the organizer

🎫 For Attendees

- 100% guaranteed authentic ticket — NFT ownership is irrevocable proof of legitimate purchase
- Protected from overpaying: smart contract caps resale price at 150% of face value
- Seamless wallet-based entry — no app download needed, just connect MetaMask at the gate
- Ticket as a digital collectible: post-event NFT unlocks exclusive content & loyalty rewards
- Transparent price history — view complete resale chain to assess fair market value before buying

Future Scope — AI Integration

Combining Blockchain's transparency with Machine Learning's predictive intelligence

Random Forest + LSTM Classifier

AI-01

Anti-Scalping Bot Detection

- Input features: purchase velocity, IP clustering, wallet age, tx frequency
- Detects bot patterns: multiple buys in <1 s, sequential wallet addresses
- On-chain blacklist automatically updated via Chainlink oracle from ML output
- False positive rate < 2% using precision-recall tuned threshold (F1 = 0.94)

XGBoost Regression + ARIMA

AI-02

Dynamic Demand Pricing Oracle

- Predicts fair market resale price using: artist popularity, weather, time-to-event
- On-chain price oracle feeds ML output directly into the resale cap smart contract
- ARIMA time-series for early demand surge detection (7-day horizon)
- Reduces organizer revenue loss from underpricing initial ticket releases

Isolation Forest + Graph Neural Net

AI-03

Fraud Anomaly Scoring

- Graph-based analysis of wallet-to-wallet ticket transfer patterns on-chain
- Isolation Forest detects outlier resale velocity (e.g., 50+ resells in 24 hrs)
- Risk score (0-100) computed per wallet, flagged for human review above threshold 70
- GNN captures cartel-like scalper networks that evade single-wallet detection

Block Ticket — The Future of Trust in Events

- 🔗 ERC-721 NFT tickets eliminate counterfeiting at the protocol level
- ⏱️ Time-based cryptographic QR codes make screenshot fraud technically impossible
- 💸 Smart contract royalties restore revenue control to event organizers
- 🤖 Integrated ML pipelines proactively neutralize bot scalping at purchase time
- 🌐 Fully decentralized — no single point of failure, censorship-resistant by design

Connect With Us

👤 Team Name: Neural Nexus

📧 Members :
- **Prasanna Nadrajan**
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🐙 GitHub: **https://github.com/Prasanna-Nadrajan/Blockchain-based-ticket-booking-system-proto**

Thank You · Q&A